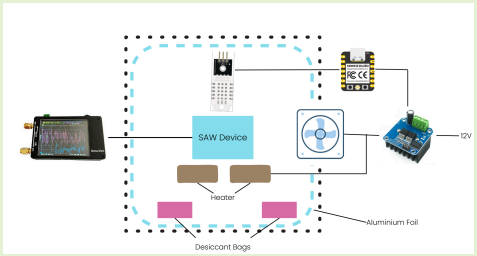
Fabrication and Temperature Characterization of a One-Port SAW Resonator on 128°YX-cut $LiNbO_3$

Wu Li Min[1], Liew Wei Shen[1], Gary Tan Jia Hoong[1], Tang Wei Zhi[1]
[1]*Department of Electrical and Computer Systems Engineering, Monash University Malaysia.*

***Abstract*—This study investigates the fabrication and temperature sensitivity of one-port surface acoustic wave (SAW) resonators on 128°YX-cut lithium niobate ($LiNbO_3$) and aluminium nitride (AlN) substrates. The devices were fabricated using thermal evaporation and contact photolithography, followed by surface characterisation using field-emission scanning electron microscopy (FESEM), atomic force microscopy (AFM), and reflectometry. Their S_{11} responses were measured using a vector network analyser (VNA) within a custom-built, proportional–integral (PI)-controlled thermal chamber. The $LiNbO_3$ resonator exhibited a resonant frequency of 129.6 MHz, within 0.3% of the design value, and a temperature sensitivity of −6.2 ± 1.0 kHz/°C over 29.2–40.6 °C, corresponding to a temperature coefficient of frequency (TCF) of −48 ± 7 ppm/°C (R^2 = 0.86). The measured TCF magnitude was lower than the literature value of approximately 75 ppm/°C, potentially due to thermal lag between the chamber air and the insulated device. The chamber heating response times ranged from 196 to 407 s, while cooling from 39.8 to 30.0 °C required approximately 770 s. In contrast, the AlN device exhibited no measurable resonance within the investigated frequency range, potentially due to weaker electromechanical coupling, electrode mass loading, and fabrication-related losses. These findings demonstrate the potential of $LiNbO_3$-based SAW resonators for passive temperature sensing while highlighting the importance of fabrication quality, thermal measurement accuracy, and device packaging.**



***Index Terms*—aluminium nitride, lithium niobate, one-port resonator, photolithography, surface acoustic wave (SAW), temperature sensor.**

I. INTRODUCTION

Temperature sensitivity materials are widely used in MEMS devices. Many industrial, medical and environmental systems need sensors that are compact, low-cost and able to work without a battery. Surface acoustic wave (SAW) devices are one example that fit the requirements. The responses of a SAW device are governed by thermo-elastic and thermo-optic properties of the piezoelectric substrate. Therefore, SAW devices serve as reliable passive temperature sensors for harsh and inaccessible environments [1].

Lithium niobate ($LiNbO_3$) and aluminium nitride (AlN) is widely utilized due to its high electromechanical coupling coefficient and robust propagation properties [2]. Change in ambient temperature and relative humidity can induce physical expansion and alters the elastic and piezoelectric constants of the substrate. This temperature-dependent characteristic can affect the acoustic wave velocity, resulting in a predictable shift in the devices' resonant frequency.

This paper shows the fabrication process of the one-port SAW resonators through thermal evaporation and contact photolithography. The substrates are 128°YX-cut $LiNbO_3$ and Aluminium Nitride (AlN), and characterised by FESEM, AFM and reflectometry. The input reflection coefficient (S_{11}) of the SAW resonator is measured with NanoVNA enclosed in a custom built heat chamber. A XIAO-ESP32 C3 was used to log the NanoVNA sweeping data into a CSV file, alongside with a custom-built web-based data monitor. The goal is to compare the temperature sensitivity, repeatability and response of the two substrates. The main contributions are: (i) a low-cost fabrication and packaging of the SAW devices; (ii) a safety-limited PI-controlled test chamber; and (iii) measured sensitivity of a 129.6 MHz $LiNbO_3$ resonator compared with AlN at a low temperature environment.

II. METHODOLOGY

A. Substrate Preparation and Al Deposition

Diced 128°YX-cut $LiNbO_3$ substrates were cleaned ultrasonically in acetone, isopropyl alcohol (IPA) and deionised (DI) water, then dried with N_2. Aluminium (Al) was deposited by thermal evaporation from a tungsten basket at a base pressure of about 5×10^{-5} mbar and a heating current of about 60 A. The Al pellets were briefly etched beforehand to remove the native oxide. The target film thickness was 200 nm.

B. Photolithography and Device Design

The resonator pattern was defined by contact photolithography and wet etching, as shown in Fig. 1. After a dehydration bake at 105 °C for 1 minute, S1805 positive photoresist was spin-coated at 2000 rpm for 20 s (acceleration 1000 rpm/s), giving a nominal thickness of about 200 nm. The resist was soft-baked at 105 °C for 1 minute and exposed through a chrome mask to 365 nm UV light for 30 s. It was developed for 3–5 min until the pattern was fully cleared under inspection, rinsed in DI water and hard-baked at 115 °C

for 1 min. Each bake was followed by 5 min of cooling. The exposed Al was then etched for 3–7 min , with the etching monitored by eye. The remaining resist was stripped in acetone with 10 s of sonication, followed by an IPA rinse and N_2 drying.

Each one-port SAW resonator consisted of an interdigital transducer (IDT) positioned between two reflector gratings. (Fig. 2). The mask defined a design wavelength of $\lambda = 30$ µm, corresponding to an electrode pitch of $\lambda/2 = 15$ µm and a finger width of $\lambda/4 = 7.5$ µm. Optical microscopy and AFM confirmed a fabricated pitch of about 15.0 µm and a finger width of about 7.5 µm. With a SAW velocity of about 3900 m/s for 128°YX-cut $LiNbO_3$, the expected resonant frequency was $f_0 = v/\lambda \approx 130$ MHz.

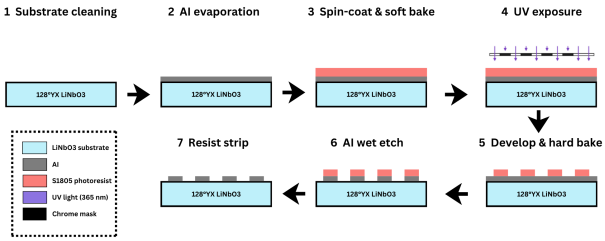


Fig. 1. Fabrication process flow of the one-port SAW resonator on 128°YX-cut $LiNbO_3$.

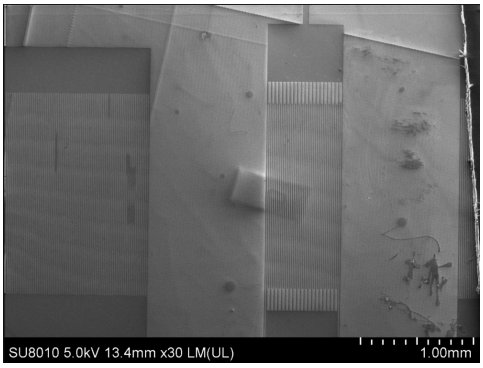


Fig 2. FESEM image of the fabricated device (design wavelength $\lambda = 30$ µm).

C. Material Characterisation

Electrode morphology was examined by field-emission scanning electron microscopy (FESEM, Hitachi SU8010) at 5 kV. The electrode step height and surface roughness were measured by atomic force microscopy (AFM, Bruker MultiMode 8) in tapping mode over a 50 × 50 µm² area. The Al film thickness was also measured with a Filmetrics F20 reflectometer.

D. Packaging and Temperature Test Chamber

Each SAW die was mounted on a printed circuit board (PCB), with its bond pads connected to an SMA connector using copper tape instead of conventional wire bonding. A foam spacer and a 3D-printed cover were used to protect the device.The temperature test chamber was constructed from a sealed plastic storage box lined with aluminium foil (Fig. 3). Two heating elements were connected in parallel and powered by a 12 V supply through a BTS7960 (HW-039) motor driver.

An ESP32-C3 microcontroller monitored temperature and humidity using a DHT22 sensor positioned near the device, with readings taken every 2 s. The heaters initially operated at full power until the chamber temperature approached within approximately 1 °C of the setpoint, after which a proportional–integral–derivative (PID) controller regulated the heating power. A DC fan was activated when the chamber temperature exceeded the setpoint to assist cooling.

Temperature and humidity measurements were transmitted to a laptop via USB, allowing the chamber to be monitored and controlled through serial commands.For safety, the system was powered through a 13 A fused plug, and the maximum operating setpoint was limited to 54 °C. If the chamber temperature reached 55 °C or a sensor failure was detected, the heaters were automatically disabled and the fan was activated. These safety mechanisms were verified before experimental measurements.

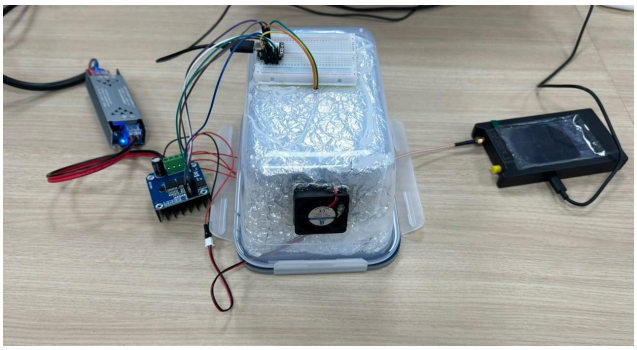


Fig. 3. Test chamber and measurement setup.

E. Measurement Procedure

The S_{11} response of the device was measured with a NanoVNA-H4 vector network analyser controlled by NanoVNA-Saver. The frequency was swept from 127 to 133 MHz with 101 points (60 kHz step), using an IF bandwidth of 100 Hz and averaging over five sweeps with the two largest outliers discarded.The chamber temperature was increased from about 29.2 °C to 40.5 °C, the maximum reachable, in steps of about 2 °C. At each step, the temperature was held until it was stable within ±0.3 °C, and the heaters were switched off during the S_{11} sweep to avoid electromagnetic interference from the PWM drive. The temperature and humidity at each measurement were taken from the DHT22 log at the time each S_{11} file was saved.The resonant frequency f_0 was found by fitting a smooth resonance curve (a Lorentzian) to the $|S_{11}|$ dip, which locates the dip centre between the 60 kHz frequency points with an uncertainty of about ±5 kHz. The sensitivity S was taken as the slope of a straight-line fit of f_0 against temperature, and the TCF was obtained by dividing S by f_0 at the lowest temperature[4]:

$$S = \frac{\Delta f_0}{\Delta T} \quad (1)$$

$$TCF = \frac{S}{f_0} \times 10^6 \quad (2)$$

where S is in kHz/°C and TCF is in ppm/°C.

E. Data Collection Procedure

A custom built python file was created to automate the data collecting process. For each temperature, the logger waits until the chamber holds the setpoint, switches the heater off, and takes 5 sweeps on the NanoVNA. It logs f_0, peak dB, temperature and RH to saw_log.csv, and saves the full dB-versus-frequency curves to saw_log_traces.csv. The live dashboard at localhost:8000 shows progress. After the run, python clean.py saw_log.csv rejects outlier sweeps. It outputs

per-temperature means, repeatability, and the sensitivity in kHz/°C.



Fig.4. Live Dashboard showing the data collected.

III. RESULTS AND DISCUSSION

A. Surface Characterisation

AFM (Fig. 5. (a)) showed flat Al electrodes with a step height of about 270 nm, higher than the 200 nm target because the evaporator had no thickness monitor. The finger width about 7.5 µm and period about 14.5 µm were close to the design. Edge protrusions of 20–40 nm were observed, likely from etch residue. FESEM (Fig. 5. (b)) confirmed continuous, evenly spaced fingers, but also showed small particles and a local edge defect, which can short adjacent electrodes across the 7 µm gap.

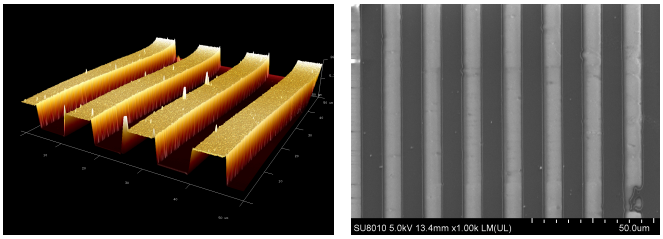


Fig. 5. (a) AFM 3D height map (50 × 50 µm²) of the Al IDT electrodes.(b) FESEM image (×1,000) of the IDT fingers on 128°YX-cut $LiNbO_3$.

B. Electrical Characterisation

The $LiNbO_3$ resonator showed a clear resonance at f_0 = 129.6 MHz (Fig. 6), within 0.3 % of the expected 130 MHz, corresponding to an effective SAW velocity of about 3888 m/s. The return loss at resonance was only 0.71 dB, indicating weak coupling to the 50 Ω system. This is attributed to the copper-tape contacts and the electrode defects observed in Fig. 6 (a).In contrast, the AlN resonator exhibited no clearly distinguishable resonance within the investigated frequency range. As a result, its resonant frequency and temperature sensitivity could not be reliably extracted. The DC current–voltage (I–V) characteristics were measured using a DC power supply and a 1 kΩ series resistor. At supply voltages of 2.5, 3.0, 3.75, 4.5 and 5.0 V, the measured device voltages were 0.03, 0.102, 0.210, 0.318 and 0.39 V, respectively. The corresponding currents were calculated as 2.47, 2.898, 3.540, 4.182, 4.61 mA using Ohm's law, as shown in Fig. 6 (b).

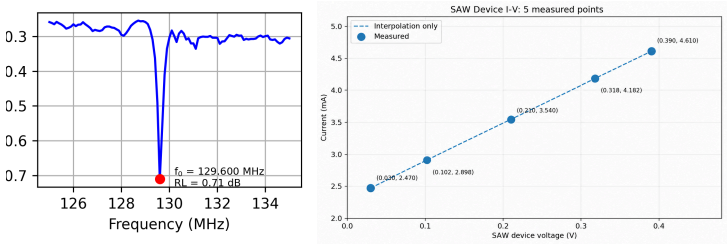


Fig.6. (a) Measured S_{11} of the $LiNbO_3$ resonator at room temperature. (b) DC current–voltage characteristics of the SAW device obtained using a 1 kΩ series resistor.

C. Temperature Sensitivity

As the temperature increased from 29.2 °C to 40.6 °C, the maximum temperature the chamber could reach, the resonance shifted to lower frequency (Fig. 6b), and f_0 decreased by 78 kHz. A linear fit (Fig. 6a) gave a sensitivity of S = −6.2 ± 1.0 kHz/°C (R^2 = 0.86), corresponding to TCF = −48 ± 7 ppm/°C. The negative TCF is expected for 128°YX-cut $LiNbO_3$, because its elastic constants soften with increasing temperature and reduce the SAW velocity. The measured magnitude is lower than the value of about −75 ppm/°C reported in the literature [ref]. This is attributed mainly to thermal lag: the DHT22 measured the chamber air, while the die was insulated by the foam spacer and 3D-printed cover, so the device temperature likely lagged the recorded value. The scatter below 31 °C also reflects the 60 kHz frequency step of the sweep, which limited the precision of f_0 extraction. The temperature range was limited to 40.5 °C because, above about 37 °C, the heaters ran at full power and the heat loss through the plastic enclosure balanced the heating power.

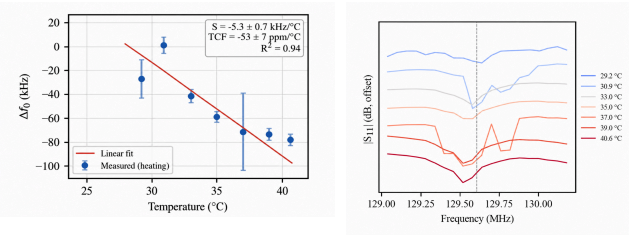


Fig. 7. Temperature response of the $LiNbO_3$ SAW resonator: (a) resonant frequency shift with linear fit (29.2–40.5 °C); (b) measured S_{11} spectra, vertically offset for clarity, with the dashed line indicating the reference resonance at 29.2 °C.

D. Repeatability, Response and Recovery

Repeatability was evaluated from repeated sweeps at fixed temperatures between 29 °C and 39 °C (Table I), with five valid sweeps at most temperatures. The weak resonance signal (0.71 dB) and measurement noise may have affected the accuracy of resonant frequency extraction.

TABLE I. Repeatability of the extracted resonant frequency at selected temperatures

Temperature (°C)	Mean f_0 (MHz)	Valid measurements
29.14	131.854	5
31.17	131.387	3
33.08	130.414	5
35.07	131.42	5
36.92	130.31	5
39.12	129.293	5

The chamber thermal response is given in Tables II and III. On heating, t_{90} was 196–407 s for 2 °C steps starting from 29.0–37.1 °C. On cooling from 39.8 °C after heating stopped, the chamber reached 35.0 °C in 340 s, 30.0 °C in 770 s and 28.9 °C in 954 s.

TABLE II. Chamber thermal response on heating

Setpoint (°C)	Initial T (°C)	Chamber t_{90} (s)	90% threshold (°C)
31	29	252	30.80
33	30.1	407	32.71
35	33.1	388	34.81
37	35	302	36.80
39	37.1	196	38.81

TABLE III. Chamber recovery time on cooling

Cooling condition	Recovery time
39.8 → 35.0°C	340.0 s
39.8 → 30.0°C	770.2 s
39.8 → 28.9°C	954.1 s
39.8 → 30.0°C (t_{90})	661.0 s

E. Discussion

The fabricated 128°YX-cut LiNbO3 resonator successfully demonstrated temperature sensing capabilities, yielding a measured temperature coefficient of frequency (TCF) of -48 ± 7 ppm/°C. While this is lower in magnitude than the theoretical expectation of -75 ppm/°C due to thermal lag between the physical die and the ambient chamber sensor, the device reliably generated surface acoustic waves. In contrast, the Aluminum Nitride (AlN) substrate failed to produce any measurable resonance response or S_{11} return loss peak at the targeted operating frequency. This functional divergence is directly tied to the fundamental material properties of the two substrates interacting with the specific fabrication defects of this batch. LiNbO3 possesses a high electromechanical coupling coefficient ($K^2 \approx 5.5\%$), which allows it to forcefully generate acoustic waves even when burdened by non-ideal conditions [3]. AlN, conversely, has a significantly lower coupling coefficient ($K^2 < 5.5\%$) [3]. Because the fabrication process lacked an evaporation thickness monitor, the deposited aluminum electrodes measured 270 nm instead of the targeted 200 nm. For the high-K^2 LiNbO3, this excess mass loading and the 20-40 nm edge protrusions caused impedance mismatching but did not completely extinguish the signal. For the low K^2 of AlN, the heavy mass loading of the thick electrodes fundamentally damped the weak acoustic wave generation [3]. While the strong piezoelectric effect of LiNbO3 could partially drive through this poorly matched interface, the weak AlN signal was entirely attenuated.

IV. FUTURE RECOMMENDATIONS

A. Thermal Measurement Precision

To eliminate the thermal lag that skewed the LiNbO3 TCF calculations, a surface-mount thermistor should be placed directly on the PCB adjacent to the SAW die, providing localized die temperature rather than ambient chamber air temperature.

B. Testing Chamber Upgrade

The custom testing chamber requires advanced thermal insulation such as thicker high-density foam to prevent the heat loss that capped the maximum testing temperature at 40.5 °C, allowing full characterization up to the targeted 55 °C.

C. Packaging And Impedance Matching

Future devices must utilize standard wire bonding to connect the die pads directly to the PCB. Additionally, an external impedance matching network like L-network or Pi-network should be designed on the PCB to maximize power transfer to the AlN substrate and lift its weak resonance out of the noise floor.

V. CONCLUSION

This project successfully demonstrated the design, characterization, and thermal evaluation of a one-port SAW resonators with a 30 µm wavelength were fabricated on 128°YX-cut $LiNbO_3$ and AlN using thermally evaporated aluminium and contact photolithography.The $LiNbO_3$ Interdigital transducer (IDT) configuration achieved a nominal resonance at 129.6 MHz, within 0.3 % of the design value, corresponding to an effective SAW velocity of about 3888 m/s. Over 29.2–40.6 °C, the resonance shifted downward with a sensitivity of -6.2 ± 1.0 kHz/°C ($R^2 = 0.86$), giving a TCF of -48 ± 7 ppm/°C. This has the expected TCF trend but is smaller in magnitude than the literature benchmark value given by (-75 ppm/°C). We attribute the difference mainly to thermal lag between the chamber-air sensor and the insulated die. Overall, the findings confirm that the SAW resonator provides a robust, passive, and highly thermal sensitive platform for temperature sensing, making it an excellent candidate for deployment in harsh or inaccessible environments where active, battery-powered electronics are impractical.

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AI DECLARATION

ChatGPT (https://chat.openai.com/) was used to assist with test planning, and language editing of author-drafted text. All measurements were performed by the authors, and all AI-assisted content was checked against the raw data and revised by the authors.

REFERENCES

[1] D. D. J. Sinhalathilakage, S. Corrie, and N. Ramakrishnan, "Advancements in wireless passive surface acoustic wave sensors and their applications: A review," Sensors and Actuators A: Physical, vol. 408, p. 117922, 2026/10/01/ 2026, doi: https://doi.org/10.1016/j.sna.2026.117922.

[2] R. Ma, W. Liu, X. Sun, S. Zhou, and D. Lin, "FEM simulation of a high-Performance 128 Y–X LiNbO3/SiO2/Si functional substrate for surface acoustic wave gyroscopes," Micromachines, vol. 13, no. 2, p. 202, 2022.

[3] M. Clement, L. Vergara, J. Sangrador, E. Iborra, and A. Sanz-Hervás, "SAW characteristics of AlN films sputtered on silicon substrates," Ultrasonics, vol. 42, no. 1, pp. 403–407, 2004/04/01/ 2004, doi: https://doi.org/10.1016/j.ultras.2004.01.034.

[4] Z. Zhou, H. Wang, and L. Lou, “Design and characterization of surface acoustic wave-based wireless and passive temperature sensing system,” Micromachines, vol. 15, no. 4, Art. no. 544, Apr. 2024, doi: 10.3390/mi15040544.

AUTHORS’ BIOGRAPHIES



Li Min Wu is a third-year student in the Bachelor of Electrical and Computer Systems Engineering at Monash University Malaysia, which he joined in 2024.



Liew Wei Sheni is a third-year student in the Bachelor of Electrical and Computer Systems Engineering at Monash University Malaysia, which he joined in July 2024.



Gary Tan Jia Hoong is a second-year student in the Bachelor of Electrical and Computer Systems Engineering at Monash University Malaysia, which he joined in March 2025.



Tang Wei Zhi is a third-year student in the Bachelor of Electrical and Computer Systems Engineering at Monash University Malaysia, which he joined in July 2024. His current interests include microelectronic design and real time embedded systems.